

Razia Sultana Jui

Research Interests: Security & Privacy | Human-Centered Computing

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Education

BRAC University Summer 2018 - Fall 2024

Bachelor of Science in Computer Science and Engineering

CGPA (Majors): 3.25/4.00

Department of Computer Science and Engineering, Dhaka, Bangladesh

Research Project Experience

HarmShield: Identifying and Mitigating Harms in LLM and VLLM-Based Assisting Systems

LLM and VLLM Safety, Trauma-Informed Design, Vulnerable system 2026 - Present

- Investigating how widely deployed LLM-based agents respond to user disclosures of trauma,

partner abuse, and accident crises, with a focus on failure modes that risk

re-traumatization,

- unsafe advice, or mishandled escalation.

- Designing a structured framework grounded in safety, trustworthiness, choice, collaboration,

and empowerment principles and translating these into measurable criteria for AI system

behavior, which assists victims in the proper way.

Empirical Study of Multimodal Hate Meme Detection: Can VLMs See What Text-Only Models Miss?

Multimodal ML, VLMs, Content Moderation, Online Harms 2025 - Present

- Curated a 117-meme benchmark of visual memes content whose text is superficially benign yet

hateful when image and text are interpreted together sourced from Facebook Hateful Memes and M3 datasets.

- Benchmarked 14+ open-source VLMs across five model families and conducted a systematic definition-free vs hate-definition-conditioned study revealing that prompt framing causes

F1 swings up to +0.637 and best overall: Gemma3-27B at F1 = 0.898.

Identifying Hate Speech in Bangla Language Text Using Natural Language Processing

Undergraduate Thesis - NLP, Deep Learning, Hate Speech Detection, Bangla Text

Classification 2024 - 2025

- Completed undergraduate thesis under the supervision of Mr. Annajiat Alim Rasel and developed

and compared deep learning models leveraging BiLSTM, CNN-LSTM, and transformer-based architectures (mBERT, XLM-R) for hate speech detection in Bangla.

- Designed a preprocessing and normalization pipeline for Bangla text that improved classification performance and robustness across domains.

Professional Projects

Gift Cards Zone Website Development | WordPress, PHP, CSS | November 2020

- Developed a full-stack e-commerce web application for a multi-purpose retail shop.
- Implemented subscription-based reviews, sales tracking, and an admin recommendation module.

Academic Projects

Fire and Smoke Detection System using Arduino | C++, Arduino Uno, DHT22, MQ2 | Summer 2022

- Designed an automated safety system to detect fire, smoke, and gas leaks.
- Integrated DHT22 sensors for temperature and humidity monitoring, displaying live data on an LCD.

Heart Disease Prediction using Machine Learning | Python, Scikit-learn, Pandas, NumPy | Fall 2022

- Developed a diagnostic model on a Kaggle dataset of 1,026 entries using KNN, Decision Tree, and Logistic Regression.
- Achieved 93% accuracy with MinMax and Standard Scaler normalization.

Real Estate Management System | Django, Python, HTML, CSS, JavaScript | Spring 2021

- Built a full-stack web platform for property trading with a dynamic 48-hour bidding system.
- Engineered role-based access, secure property listings, price filtering, and user verification.

Highlighted Academic Courses

- CSE-423 Computer Graphics
- CSE-425 Neural Networks
- CSE-420 Compiler Design
- CSE-360 Computer Interfacing

Technical Skills

Languages: Python, Java, C, C++, MATLAB, JavaFX, Bash, Assembly, Swing

ML / NLP: PyTorch, scikit-learn, pandas, NumPy, BERT, RAG, LLM, VLLM

Databases: Oracle, MySQL, SQLite, MongoDB

OS: Linux (Ubuntu, Kali), Windows

Version Control: Git, GitHub

Frameworks: Django, Node.js, React, JavaFX

Web: HTML, CSS, PHP, Bootstrap, WordPress, JSON, XML
Compiler Tools: Flex, Bison, EMU 8086
Microcontrollers: Arduino, ATmega 32, Atmel Studio, Proteus 832
Writing: LATEX, Beamer, Overleaf
Other: NS3, iGraphics, Cisco Packet Tracer, OpenGL

References

Mr. Annajiat Alim Rasel - annajiat@bracu.ac.bd
Dr. Muhammad Iqbal Hossain - iqbal.hossain@bracu.ac.bd